

Lab2

Lab2_dp

Audio_Codec_Wrapper

clk
reset_n
ctrl
ac_mclk
ac_adc_sdata(In)
ac_dac_sdata(Out)
ac_bclk
ac_lrclk
scl
sda

Clock_Wiz_1

clk
reset_n
clk_out1
clk_out2
resetn

sim_live
ready
sw
L_bus_in
R_bus_in
L_bus_out
R_bus_out

sim_live
switch(3)

Counter (unsigned)

10
3FF
write_cntr

Compare

Mux

10
exSel

Register

clk
D
Q
incoming

sign2Unsign

current

Mux

1
0
exSel

BRAM_SDP

WRADDR
Din
wrENB

clk
reset_n
btn

clk
reset_n
btn

Video

tr_volt
tr_time

ch1
ch2
ch1_enable
ch2_enable

row
column

10
10

ch1
ch2
ch1_enable
ch2_enable

row
column

10
10

ch1.from_bram

10
10
row

Compare E

ch1

ready

flag register

set

Q

clear

flagQ

flagClear

Lab2_cu

clk
reset_n

sw
cw

3
3

sw
cw

L_Bus Out
R_Bus Out

16
16

L_Bus Out
R_Bus Out

exLBus

16

exLBus

exRBus

16

exRBus

tr_volt

16

tr_volt

tr_time

16

tr_time

flagQ

flagClear

flagClear

(320, 105) → 320
 col, Row
 220 ↔ 512
 512, 105

if ~~sw~~ = 0 → Row = 20

if ~~sw~~ = 1023 → Row = 420
 512

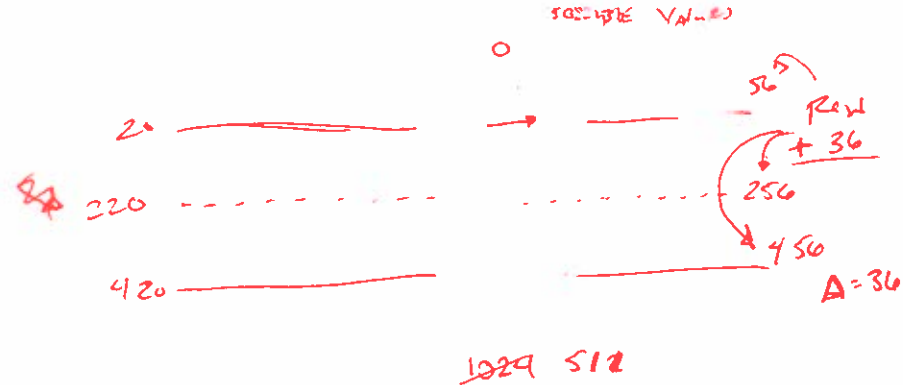
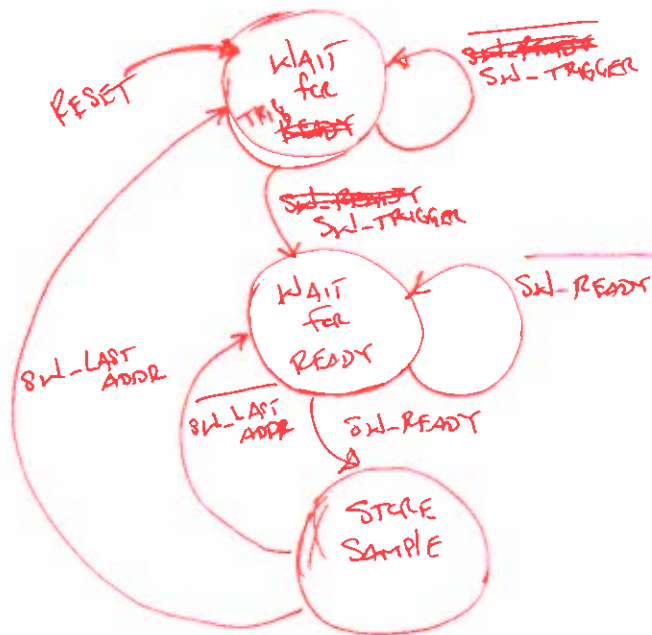
if ~~sw~~ = 512 → Row = 220 (260 + 292) = 512

if ~~sw~~ = 1023 → Row = 420 712 - 292 = 420

Row 0

- WHEN TRIGGER PANT ME-
 - CIA TO START CONTROL
 - BEGINNING TO RECORD
 SAMPLES TO BRAM

Control unit



STATE

RESET
 ↓
 WAIT for TRIGGER
 ↓
 WAIT for READY
 ↓
 STORE SAMPLE

CW-Counter-Control

00 RESET
 01 RESET
 10 Hold Count up
 11 TIED TO RESET-1
 CTRL INPUT

STATE

RESET

WAIT FOR TRIGGER

WAIT FOR READY

STORE SAMPLE

CW

CW-Counter-Control ← "00"
 CW-WRITE-EN ← '0'

CW-Counter-Control ← "00"
 CW-WRITE-EN ← '0'

CW-Counter-Control ← "10"
 CW-WRITE-EN ← '0'

CW-Counter-Control ← "11"
 CW-WRITE-EN ← '1'

